CHAPTER 9

PLANT GROWTH AND DEVELOPMENT

CHAPTER ESSENCE AT A GLANCE:
Hormonal signals and environmental triggers coordinate plant morphogenesis, cell differentiation, and growth kinetics.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Growth Kinetics and Mathematical Models:** Plant growth exhibits geometric and arithmetic kinetics: Arithmetic growth produces a linear curve (Lt = L0 + rt) where only one daughter cell retains dividing capability; Geometric growth displays a characteristic sigmoid (S-shaped) curve ($W1 = W0\ e^{r}t$) progressing through initial lag phase, rapid exponential log phase, and terminating in a stationary phase due to limited nutrient availability.

• **Differentiation, Dedifferentiation, and Redifferentiation:** Differentiation occurs when meristematic cells mature to perform specialized functions (e.g., tracheary elements losing protoplasm). Dedifferentiation occurs when living differentiated cells regain division capacity (e.g., formation of interfascicular cambium and cork cambium from parenchyma). Redifferentiation occurs when dedifferentiated cells lose division ability again to form mature secondary tissues.

• **Plant Growth Promoters: Auxin, GA, and Cytokinins:** Phytohormone promoters stimulate morphogenesis: Auxins (IAA) maintain apical dominance, initiate root formation in cuttings, and synthetic 2,4-D acts as a selective dicot weedicide; Gibberellins (GA3) promote bolting in rosette plants, internodal elongation in sugarcane, and malting in brewing; Cytokinins (zeatin, kinetin) promote cell division, overcome apical dominance, and delay leaf senescence.

• **Plant Growth Inhibitors: Ethylene and Abscisic Acid:** Phytohormone inhibitors regulate stress and senescence: Ethylene is a unique gaseous hormone promoting fruit ripening, abscission, and feminine flower induction in cucumbers, characterized by a respiratory climacteric burst; Abscisic Acid (ABA) acts as the plant stress hormone, inducing rapid stomatal closure during drought and enforcing seed dormancy, acting antagonistically to Gibberellins.

ESSENTIAL TAXONOMICAL FRAMEWORKS & PROCESSES

$L_t = L_0 + rt$ [Arithmetic Growth: Constant rate producing a linear curve]

Linear Arithmetic Root Elongation Model

$W_1 = W_0 e^{rt}$ [Geometric Growth: Sigmoid S-curve with lag, log, stationary]

Exponential Geometric Growth Equation

$\text{Promoters} = \text{Auxin (Apical Dominance)} + \text{GA (Bolting)} + \text{Cytokinin (Cell Division)}$

Morphogenetic Plant Growth Promoting Hormones

$\text{Inhibitors} = \text{Ethylene (Ripening Gas)} + \text{ABA (Stomatal Closure Stress Hormone)}$

Senescence and Abscission Regulatory Regulators

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Photoperiodism, Vernalisation, and Seed Dormancy sections were completely deleted under the rationalized syllabus.

• **Alert:** Auxin promotes apical dominance (inhibiting lateral bud growth), whereas Cytokinin counteracts apical dominance (stimulating lateral bud outgrowth).

• **Alert:** Abscisic acid (ABA) is antagonistic to Gibberellic acid (GA): ABA enforces seed dormancy, whereas GA breaks dormancy and promotes seed germination.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What is plasticity in plants? Provide two examples of plants displaying heterophylly as an expression of plasticity. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Plasticity is the capacity of plants to follow different developmental pathways in response to environment or life phases to form different kinds of structures. **[1 Mark]**

• Examples of heterophylly: Larkspur, Cotton, or Coriander (different leaf shapes in juvenile vs mature stages) and Buttercup (distinct leaf shapes in aerial vs submerged aquatic conditions). **[1 Mark]**

2. Distinguish between dedifferentiation and redifferentiation with a botanical tissue example for each. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Dedifferentiation: The phenomenon where living, fully differentiated cells regain the capacity of cell division under specific conditions (e.g., formation of interfascicular cambium or cork cambium from parenchymatous cells). **[1 Mark]**

• Redifferentiation: The process where cells produced by dedifferentiated meristems lose their division capacity again to mature into specialized secondary tissues (e.g., secondary xylem, secondary phloem, or cork). **[1 Mark]**

3. Explain three important physiological functions of Auxins and their practical applications in agriculture and horticulture. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Apical dominance: Growing apical bud inhibits the growth of lateral axillary buds; decapitation of shoot tips removes auxin source, promoting bushy lateral branching in tea plantations. **[1 Mark]**

• Root initiation: Auxins (IBA, NAA) promote adventitious root initiation in stem cuttings, widely used in vegetative propagation. **[1 Mark]**

• Weedicide action: Synthetic auxin 2,4-D is widely used as a selective herbicide to kill broad-leaved dicot weeds without harming monocot grass crops. **[1 Mark]**

4. Describe three major physiological responses induced by Gibberellins in plants, highlighting the phenomenon of bolting. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Bolting: Promotes internode elongation just prior to flowering in rosette plants like beet, cabbage, and lettuce. **[1 Mark]**

• Stem elongation: Increases stem length in sugarcane, increasing sugar yield by up to 20 tonnes per acre. **[1 Mark]**

• Malting and fruit shape: Accelerates malting process in brewing industries and elongates and improves shape of fruits like apples. **[1 Mark]**

5. Why is Abscisic Acid (ABA) termed the stress hormone in plants? Explain its role during water deficit conditions. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Stress hormone role: ABA stimulates various protective responses in plants against diverse environmental stresses such as drought, high salinity, and cold. **[1 Mark]**

• Stomatal closure: During water stress, ABA synthesized in roots and leaves triggers rapid potassium efflux from guard cells, causing loss of turgor and closure of stomata. **[1 Mark]**

• Transpirational conservation: Closing stomatal pores dramatically minimizes transpirational water loss, preventing cellular desiccation. **[1 Mark]**

6. Discuss the unique physiological role of Ethylene as a gaseous plant hormone, detailing its effects on fruit ripening and respiratory climacteric. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Ethylene is a simple gaseous phytohormone synthesized in large amounts by tissues undergoing senescence and ripening fruits. **[1 Mark]**

• Fruit ripening: Promotes rapid ripening in fruits like apples, bananas, and tomatoes, softening fruit walls and converting starch to sugars. **[1 Mark]**

• Respiratory climacteric: Induces a sharp, dramatic increase in the rate of cellular respiration during fruit ripening termed the respiratory climacteric. **[1 Mark]**

7. Compare arithmetic growth with geometric growth using their governing mathematical equations, graphic curve shapes, and cellular examples. [Score: 4 Marks]

Answer & Key Valuation Steps:

• Mathematical equations: Arithmetic growth is expressed as Lt = L0 + rt; Geometric growth is expressed as W1 = W0 * e^(rt). **[1 Mark]**

• Graphical curve: Arithmetic growth yields a linear straight line curve with constant growth rate; Geometric growth yields an S-shaped sigmoid curve featuring lag, log, and stationary phases. **[1 Mark]**

• Cellular division behavior: In arithmetic growth, following mitotic division, only one daughter cell continues to divide while the other differentiates; in geometric growth, both daughter cells retain dividing capability. **[1 Mark]**

• Biological examples: Arithmetic growth is exhibited by elongating root and shoot tips; Geometric growth is exhibited by developing embryos and unicellular populations in culture. **[1 Mark]**